CHAPTER 16

Out-of-Syllabus Quick Sheets

ARP · DHCP · ICMP · UDP · SMTP / FTP · framing · bridging · OSI layers · port numbers (one revision page each)

QUICK TRICK

These are **not named** in the GATE 2027 syllabus, so do not drill them. But they can appear as **an option or context** inside IPv4, Ethernet or socket questions. One short sheet each is enough.

16.1 ARP: IP → MAC on the same LAN

● **LEVEL 1 · BEGINNER**

- Host needs the MAC of an IP on its LAN. It **broadcasts** an ARP request (dest MAC FF:FF:FF:FF:FF:FF). The owner replies with a **unicast** ARP reply. The answer is cached.
- For a destination **off the LAN**, the host ARPs for the **default gateway's** MAC, not the far host's.
- ARP travels directly in an Ethernet frame (type 0x0806). It is a layer-2/3 helper protocol.

16.2 DHCP: automatic IP configuration

● **LEVEL 1 · BEGINNER**

```
Client (no IP yet)                                  Server
  DISCOVER  (broadcast 0.0.0.0 -> 255.255.255.255) ----->
  <----- OFFER   (proposed IP, mask, gateway, DNS, lease)
  REQUEST   (accept this offer, broadcast)        ----->
  <----- ACK     (confirmed, lease starts)            "D O R A"
```

Runs over **UDP**: server port **67**, client port **68**. Gives IP, subnet mask, default gateway, DNS server, and a **lease time**. A relay agent forwards across routers.

16.3 ICMP: error and query messages for IP

● **LEVEL 1 · BEGINNER**

Type	Message	Use
0 / 8	Echo reply / request	**ping**
3	Destination unreachable	No route, port closed, "fragmentation needed" (DF set)
11	Time exceeded	TTL = 0; basis of **traceroute**
5	Redirect	Better gateway exists
12	Parameter problem	Bad header field

ICMP rides inside IP (protocol number 1). An error message carries the IP header plus the first 8 bytes of the offending datagram. No ICMP error is sent about another ICMP error.

16.4 UDP (header recap)

● **LEVEL 1 · BEGINNER**

8 bytes: source port, destination port, length, checksum (16 bits each). Connectionless, no ordering, no retransmission, no congestion control. Checksum covers a pseudo-header with the IPs. Used by DNS, DHCP, RIP, streaming.

16.5 Email and file transfer

● LEVEL 1 · BEGINNER

Protocol	Port	Role
SMTP	25	Push mail from sender to server and between servers (text commands: HELO, MAIL FROM, RCPT TO, DATA, QUIT)
POP3	110	Download mail to the client (usually deletes on server)
IMAP	143	Manage mail on the server (folders, sync)
FTP	20 / 21	**Two connections**: control on 21, data on 20. Control is "out-of-band"
SSH / Telnet	22 / 23	Remote login (SSH is encrypted)

16.6 Framing (data link)

● LEVEL 2 · EXAM

- **Character count:** first byte gives the frame length (one error ruins every later frame).
- **Byte stuffing:** flag byte marks the frame boundary; if the flag occurs in data, an **escape** byte is inserted before it.
- **Bit stuffing (HDLC):** flag = 01111110. After **five consecutive 1s** in data, the sender inserts a 0.

```
Example   : 011011111101111101  ->  0110111110101111101   (a 0 added after every five 1s)
Receiver  : after five 1s, remove the following 0.
```

16.7 Bridges and spanning tree

● LEVEL 1 · BEGINNER

A **transparent bridge** learns MAC → port from source addresses (see Chapter 7). With loops in the LAN, **Spanning Tree Protocol** elects a root bridge (lowest ID), keeps the least-cost path to it on each bridge (root port), and **blocks** extra ports so the active topology is loop-free.

16.8 OSI layers at a glance

● LEVEL 1 · BEGINNER

#	OSI layer	Job	PDU	TCP/IP layer
7	Application	User services (HTTP, DNS, SMTP)	Data	Application
6	Presentation	Format, encryption, compression	Data	Application
5	Session	Dialogue control	Data	Application
4	Transport	Process-to-process, reliability	Segment	Transport
3	Network	Routing, logical addressing	Packet	Internet
2	Data link	Framing, MAC, error detection	Frame	Link
1	Physical	Bits on the medium	Bit	Link / physical

16.9 Well-known ports

Port	Service	Transport
20 / 21	FTP data / control	TCP
22	SSH	TCP
23	Telnet	TCP
25	SMTP	TCP
53	DNS	UDP (TCP for large)
67 / 68	DHCP server / client	UDP
80	HTTP	TCP
110 / 143	POP3 / IMAP	TCP
443	HTTPS	TCP

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```
ARP : broadcast request, unicast reply (off-LAN -> ARP for gateway)
DHCP: Discover, Offer, Request, Ack over UDP 67/68      ICMP: ping 8/0, time exceeded 11, unreachable 3
Bit stuffing: 0 after five 1s      SMTP 25  POP3 110  IMAP 143  FTP 20/21  DNS 53  HTTP 80
```

Practice questions

● **LEVEL 1 · BEGINNER**

Q1. Is the ARP request unicast or broadcast?

Q2. Which ICMP message does ping use?

Q3. Four DHCP messages in order.

Q4. Ports for SMTP, DNS and DHCP server?

● **LEVEL 2 · EXAM**

Q5. Host A wants to send to a host on another network. Whose MAC does it ARP for?

Q6. Apply bit stuffing to 011011111101111101.

Q7. Which ICMP message type signals a TTL that reached 0?

Q8. Why does FTP use two connections?

Answers

Q	Solution
1	**Broadcast**. The reply is unicast.
2	**Echo request (type 8)** and **echo reply (type 0)**.
3	**Discover, Offer, Request, Acknowledge**.
4	**25, 53, 67**.
5	The **default gateway's** MAC.
6	**0110111110101111001** (a 0 after each run of five 1s).
7	**Time exceeded (type 11)**.
8	One **control** connection (port 21) carries commands; separate **data** connections carry files (port 20 in active mode).

Next: Chapter 17 — Master Formula Sheet and Mixed Practice.